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GPS-BASED GEO-FENCING AND TRACKING SYSTEM FOR LIVESTOCK MANAGEMENT

(Course Code:BEC7028)

A Project Report

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CHAPTER 1: INTRODUCTION

1.1 Background

Livestock farming is an important part of agriculture and rural livelihoods, particularly in areas where cattle are allowed to graze freely in open fields, forests, and large grazing areas. Cattle require regular monitoring to ensure their safety and prevent them from wandering away from the designated grazing area. However, continuously observing livestock can be difficult and time-consuming for farmers and herders, especially when the animals are spread over a large or remote area.

Traditional livestock monitoring methods mainly depend on physical observation, herding, or manually searching for missing animals. These methods require continuous human effort and may not provide immediate information about the animal's location. Cattle can cross their intended grazing boundaries, enter restricted or unsafe areas, or become lost without the owner knowing their exact location. Therefore, an automated system capable of tracking livestock and providing timely alerts can significantly improve the efficiency and reliability of livestock management.

The GPS-Based Geo-Fencing and Tracking System for Livestock Management is designed to address these challenges using modern communication and positioning technologies. The system uses a GPS module to obtain the real-time latitude and longitude of the cattle. An ESP32 microcontroller processes the GPS information and compares the cattle's current position with a predefined 50-meter geo-fence. If the cattle remains within the boundary, the system considers the situation normal. If it moves outside the boundary, the system identifies it as an alert condition.

For communication, LoRa is used to transmit the cattle's location and status to a monitoring station over a suitable distance, while the SIM808 GSM module can be used to send notifications when an alert occurs. The monitoring station displays the received information and stores important events in a PostgreSQL database for monitoring and record keeping. Overall, the proposed system aims to reduce manual monitoring, provide timely information about cattle movement, and improve the safety and management of livestock in grazing areas.

This project focuses on designing and implementing a GPS-Based Geo-Fencing and Tracking System that continuously monitors the location of humans and animals, detects boundary violations, and provides immediate notifications to improve safety and security.

1.2 Problem of Statement

Livestock, particularly cattle, are often left to graze or move freely over large areas, making continuous monitoring difficult for farmers and herders. Cattle may wander outside the intended grazing area, get lost, or enter unsafe locations without the owner being immediately aware. Traditional monitoring methods depend mainly on manual observation, which can be time-consuming and unreliable, especially in large or remote areas.

Therefore, there is a need for an automated livestock tracking and geo-fencing system that can continuously monitor the location of cattle and provide an immediate alert when they move outside a predefined boundary. The proposed system uses GPS to determine the cattle's location, ESP32 for processing, LoRa for wireless communication, and GSM for sending alerts. This system aims to improve livestock safety, reduce the need for constant manual monitoring, and help farmers quickly identify and locate cattle that leave the designated area.

1.3 Objective

1.3.1 Main Objective

- To design and implement a GPS-Based Geo-Fencing and Tracking System capable of monitoring the location of livesticks and generating real-time alerts whenever predefined geographical boundaries are crossed.

1.3.2 Specific Objectives

- To acquire accurate location information using GPS technology.
- To establish virtual geographical boundaries through geo-fencing techniques.
- To transmit real-time alerts using wireless communication technologies.

1.4 Scope and Limitations of Study

1.4.1 Scope

- Farmers can monitor the location of animals and receive immediate alerts if livestock leave designated grazing areas.
- Researchers and conservation authorities can track wildlife movement and reduce human-animal conflicts.

1.4.2 Limitations

- GPS accuracy may be affected by dense forests, buildings, and adverse weather conditions.
- Communication range depends on the selected wireless technology.
- The prototype is designed for small-scale deployment and testing.

CHAPTER 2: LITERATURE REVIEW

Kaplan and Hegarty (2006) explained the principles and applications of Global Positioning System (GPS) technology. Their work demonstrated how satellite-based navigation systems provide accurate location information for tracking and positioning applications. GPS remains one of the most reliable technologies for outdoor localization and navigation systems.

Li et al. (2020) reviewed localization techniques used in IoT systems and discussed the challenges associated with positioning accuracy, communication reliability, and energy consumption. The study identified GPS as one of the most widely adopted technologies for outdoor tracking applications and highlighted the importance of integrating localization with wireless communication technologies.

Khan et al. (2021) conducted a systematic review of location-aware schemes in the Internet of Things (IoT). The study highlighted the importance of localization technologies in safety monitoring, healthcare, transportation, and environmental applications. The authors emphasized that location-based services play a critical role in enhancing real-time decision-making and monitoring systems.

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Tobin et al. (2023) reviewed the application of geo-fencing technologies in monitoring and intervention systems. Their findings showed that geo-fencing enables automated detection of entry and exit events within predefined geographic boundaries, making it suitable for safety, health monitoring, and location-based alert systems. The study demonstrated the growing adoption of geo-fencing technology in real-time monitoring applications.

Bonilla et al. (2023) conducted a systematic review of LoRaWAN technology and its applications in IoT systems. The authors reported that LoRaWAN provides long-range communication, low power consumption, and reliable connectivity, making it highly suitable for remote monitoring and tracking applications where traditional communication infrastructures may not be available.

Shahbazian et al. (2023) reviewed machine learning-assisted localization systems for IoT applications. The study highlighted how localization technologies can be combined with intelligent algorithms to improve positioning accuracy and support future smart monitoring systems. Their work indicates the growing trend toward integrating localization and

intelligent decision-making capabilities in tracking applications.

Reddy et al. (2024) reviewed an IoT-based monitoring system utilizing LoRa technology and GPS-based geo-fencing. The proposed framework employed GPS modules, LoRa communication, and sensor networks to monitor movement and generate alerts when predefined boundaries were crossed. The study demonstrated the feasibility of combining geo-fencing and long-range communication technologies for large-area monitoring systems.

Recent studies in livestock monitoring have also demonstrated the effectiveness of combining GPS tracking, LoRa communication, and IoT technologies. These systems allow farmers to monitor animal movement, receive alerts, and improve livestock management in remote environments. Such developments indicate the practical applicability of geo-fencing and tracking systems for animal safety and monitoring.

2.1 Literature Comparison

Table 2.1: Literature Comparison of GPS-Based Geo-Fencing and Tracking Systems

Author(s) and Year	**Focus of Study**	**Key Contribution**
Kaplan & Hegarty (2006)	GPS Technology	Provided the fundamentals of GPS and satellite-based positioning systems.
Khan et al. (2021)	Location-Aware IoT Systems	Reviewed localization methods and their applications in IoT systems.
Li et al. (2020)	IoT Localization Techniques	Discussed positioning techniques, challenges, and error mitigation methods.
Tobin et al. (2023)	Geo-Fencing Applications	Studied the use of geo-fencing for automated monitoring and alerts.
Bonilla et al. (2023)	LoRaWAN and IoT Communication	Evaluated LoRaWAN for reliable long-range IoT communication.
Shahbazian et al. (2023)	Machine Learning-Based Localization	Reviewed machine learning approaches to improve localization accuracy.
Reddy et al. (2024)	LoRa-Based Monitoring Systems	Presented GPS and LoRa integration for real-time monitoring applications.

CHAPTER 3: METHODOLOGY

3.1 System Overview

The proposed GPS-Based Geo-Fencing and Tracking System is designed to monitor the real-time location of a human or animal within a predefined geographical area. The system uses a GPS module to continuously acquire location coordinates and a microcontroller to process the data. The current location is compared with a predefined geo-fence boundary stored in the system.

When the tracked individual remains inside the designated area, the system continues monitoring without generating alerts. However, if the individual crosses the geo-fence boundary, an alert notification containing location information is transmitted to the monitoring station through LoRa or GSM communication.

The monitoring station receives the information and displays the location and status of the tracked individual through a web-based dashboard or mobile application.

3.2 Block Diagram

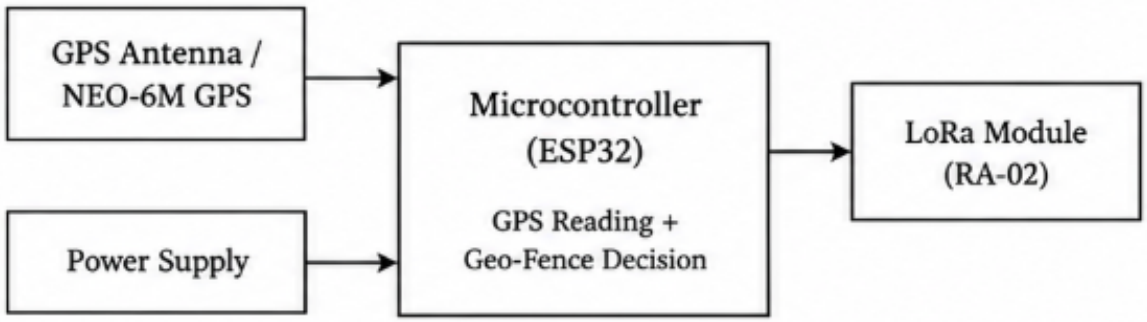


Figure 3.1: Block diagram of Cattle unit

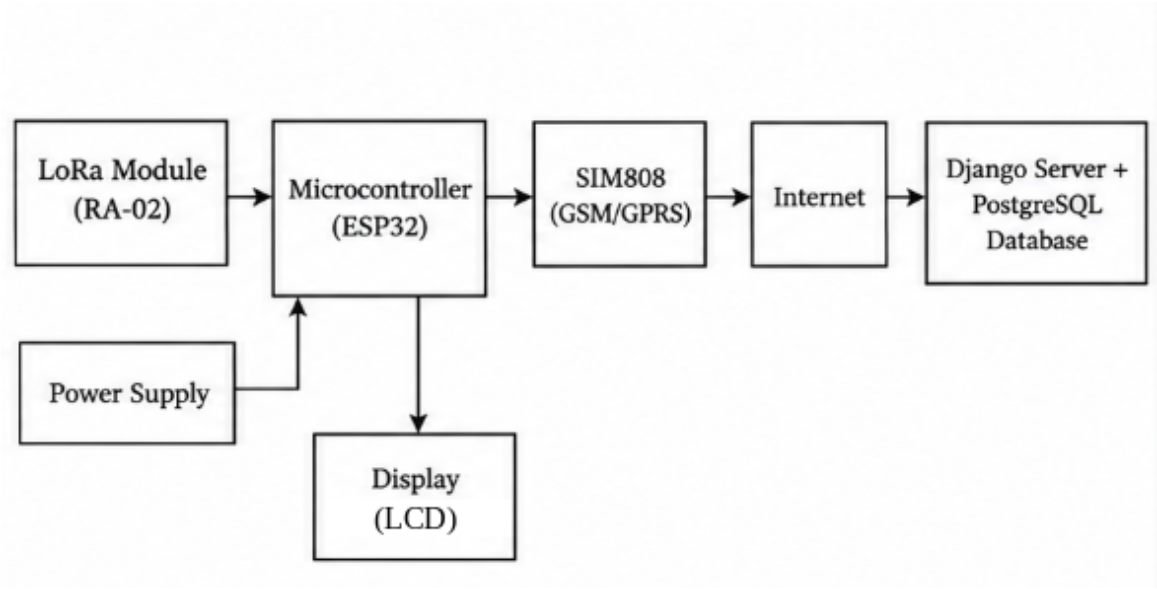


Figure 3.2: Block diagram of Monitoring Station

The block diagram consists of two main units: the Cattle Unit (Transmitter) and the Monitoring Station (Receiver). In the cattle unit, the NEO-6M GPS receives the cattle's location coordinates, which are processed by the ESP32 to determine whether the cattle is within the predefined geo-fenced area. The processed location and geo-fence status are then transmitted through the LoRa module to the monitoring station. At the monitoring station, another LoRa module receives the transmitted data and sends it to the ESP32 for processing. The information can be displayed on the OLED display, while the SIM808 GSM/GPRS module sends the data through the Internet to the Django server and PostgreSQL database, where the livestock location and alert information are stored for monitoring and future reference..

3.2.1 System Architecture

Each blocks can be described as:

Power Supply

It supplies the required electrical power to the microcontroller and all connected modules. It ensures that all the components receive stable and required power.

ESP32 Microcontroller

The ESP32 acts as the central processing unit of the system. It receives location data from the GPS module and compares the current position with the predefined geo-fence boundary

stored in memory. The ESP32 is also responsible for generating alerts, controlling peripheral devices, and managing communication with the monitoring server.

GPS Module (NEO-6M)

The GPS module continuously receives signals from GPS satellites and determines the current geographical coordinates (latitude and longitude) of the tracked human or animal. These coordinates are transmitted to the ESP32 microcontroller through serial communication.

Display

The display unit shows important information such as the current GPS coordinates, system status, geo-fence status, communication status, and alert notifications. It allows users to monitor the tracking system directly without accessing the dashboard.

Buzzer/Indicator

The buzzer serves as a local alert mechanism. Whenever the tracked subject crosses the predefined geo-fence boundary, the ESP32 activates the buzzer to indicate a boundary violation. LEDs may also be used as visual indicators for system status and alerts.

LoRa / GSM Module

The communication module enables wireless transmission of location data and alert notifications to the monitoring station. LoRa provides long-range, low-power communication suitable for remote areas, while GSM can send SMS alerts through cellular networks. When a geo-fence violation occurs, the ESP32 transmits the current location and alert message to ensure timely monitoring and response.

Database(Monitoring Server)

The monitoring server stores location data, alert logs, and movement history. It receives information from the communication module and makes it available through a web-based dashboard or mobile application. The server also allows users to configure geo-fence boundaries and monitor multiple devices simultaneously.

3.3 Hardware platform

The hardware platform of the proposed GPS-Based Geo-Fencing and Tracking System consists of several components that work together to perform location tracking, boundary monitoring, data processing, and wireless communication. The major hardware components used in the system are described below.

3.3.1 ESP-32 Microcontroller

The ESP32 serves as the main controller of the system. It processes GPS data, performs geo-fencing calculations, controls peripheral devices, and manages communication between the tracking device and the monitoring station. Its built-in Wi-Fi and Bluetooth capabilities provide additional flexibility for future enhancements.

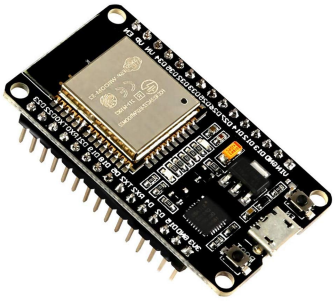


Figure 3.3: ESP32 Microcontroller

3.3.2 GPS Module (NEO-6M)

The GPS module is responsible for determining the real-time geographical location of the tracked human or animal. It receives signals from GPS satellites and provides latitude and longitude coordinates to the ESP32 for location monitoring and geo-fence analysis.

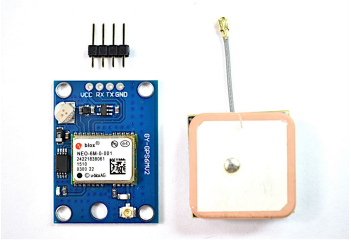


Figure 3.4: GPS Module (NEO-6M)

3.3.3 LoRa Module (SX1278)

The LoRa module enables long-range wireless communication between the tracking device and the monitoring station. It is suitable for remote and rural areas due to its low power consumption and extended communication range.



Figure 3.5: LoRa Module (SX1278)

3.3.4 LCD 16*2 Display

An LCD (Liquid Crystal Display) is an electronic display used to show text, numbers, and symbols. It is commonly used in devices like calculators, clocks, and microcontroller projects. LCDs are popular because they consume low power, are inexpensive, and can easily connect with boards such as the Arduino Uno and ESP32 to display information like sensor data and messages.

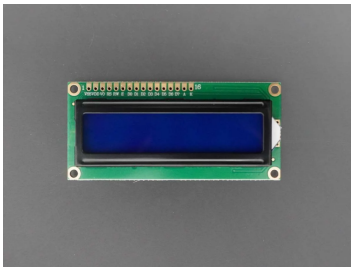


Figure 3.6: LCD 16*2 Display

3.3.5 Buzzer

The buzzer acts as an audible alert device. It is activated when the tracked subject crosses the predefined geo-fence boundary or when a warning condition is detected.



Figure 3.7: Buzzer

3.3.6 Lithium-Ion Battery

A rechargeable lithium-ion battery powers the entire tracking device. It provides portability and enables continuous operation in outdoor environments without requiring a constant power source.



Figure 3.8: Lithium-Ion Battery

3.4 Software platform

The software platform consists of the programming tools, libraries, and applications used to develop, test, and monitor the GPS-Based Geo-Fencing and Tracking System. These software components enable data processing, communication, visualization, and system management.

3.4.1 Arduino IDE

The Arduino Integrated Development Environment (IDE) is used to write, compile, and upload programs to the ESP32 microcontroller. It provides a simple interface for developing embedded applications and supports various libraries required for GPS and communication modules.

3.4.2 TinyGPS++ Library

The TinyGPS++ library is used to process GPS data received from the GPS module. It extracts useful information such as latitude, longitude, speed, and time, which are required for location tracking and geo-fencing operations.

3.4.3 LoRa Library

The LoRa library enables communication between the ESP32 and the LoRa module. It facilitates the transmission and reception of location data and alert messages over long distances.

3.4.4 Web Dashboard

A web-based dashboard is used for real-time monitoring of the tracked individual or animal. It displays location information, geo-fence status, alert notifications, and movement history in a user-friendly interface.

3.4.5 Google Maps API

The Google Maps API is integrated into the dashboard to visualize the real-time location of the tracked subject on an interactive map and display geo-fence boundaries.

3.5 Flowchart

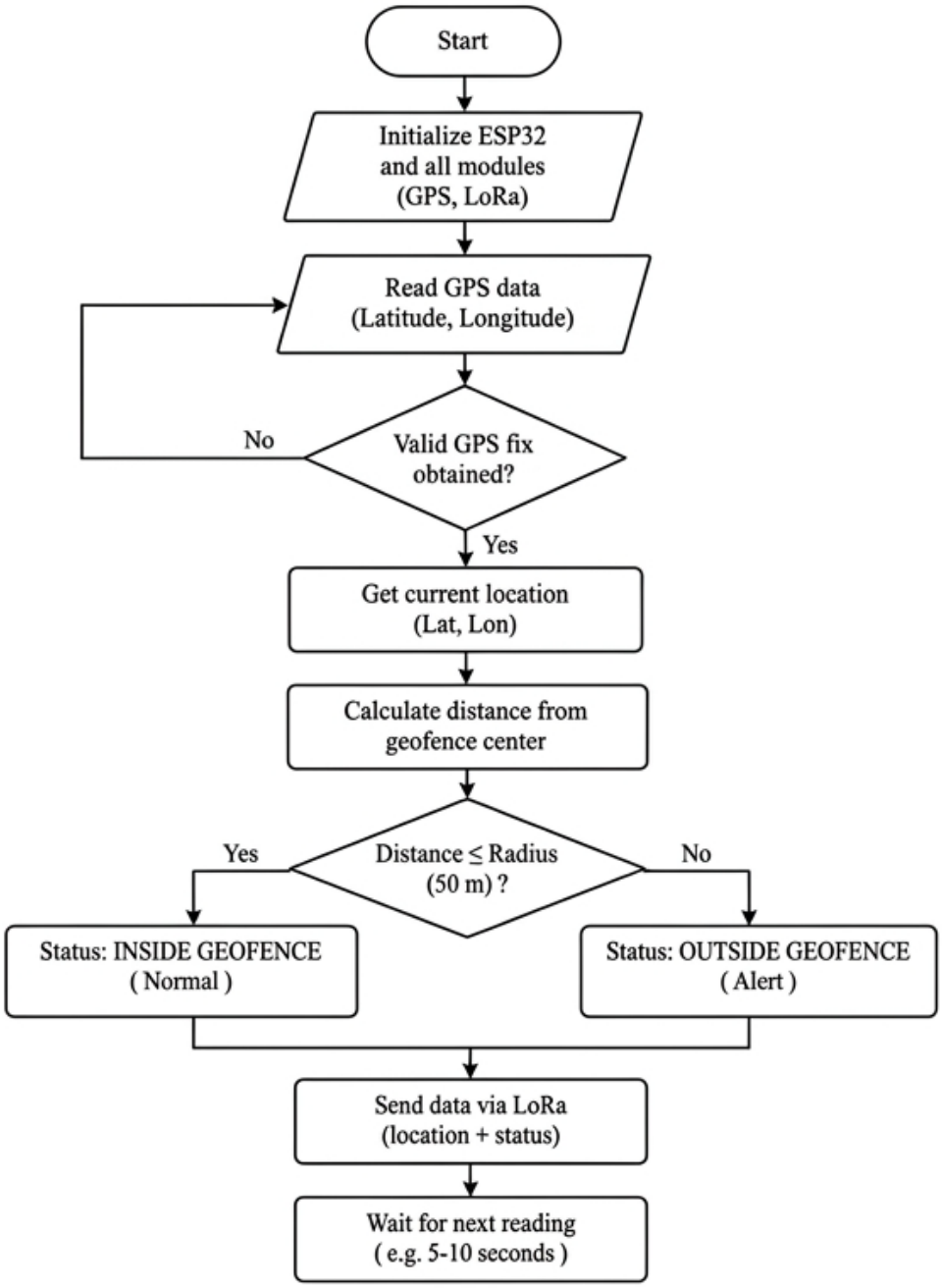


Figure 3.9: Flowchart of Cattle unit

The Cattle Unit initializes the ESP32, GPS, and LoRa modules and reads GPS data until a valid location is obtained. The system then calculates the cattle's distance from the geofence center. If the distance is within 50 meters, the status is set to "INSIDE GEOFENCE (Normal)"; otherwise, it changes to "OUTSIDE GEOFENCE (Alert)". The cattle's location and status are then transmitted to the monitoring station through LoRa. This process repeats every 5–10 seconds for continuous tracking.

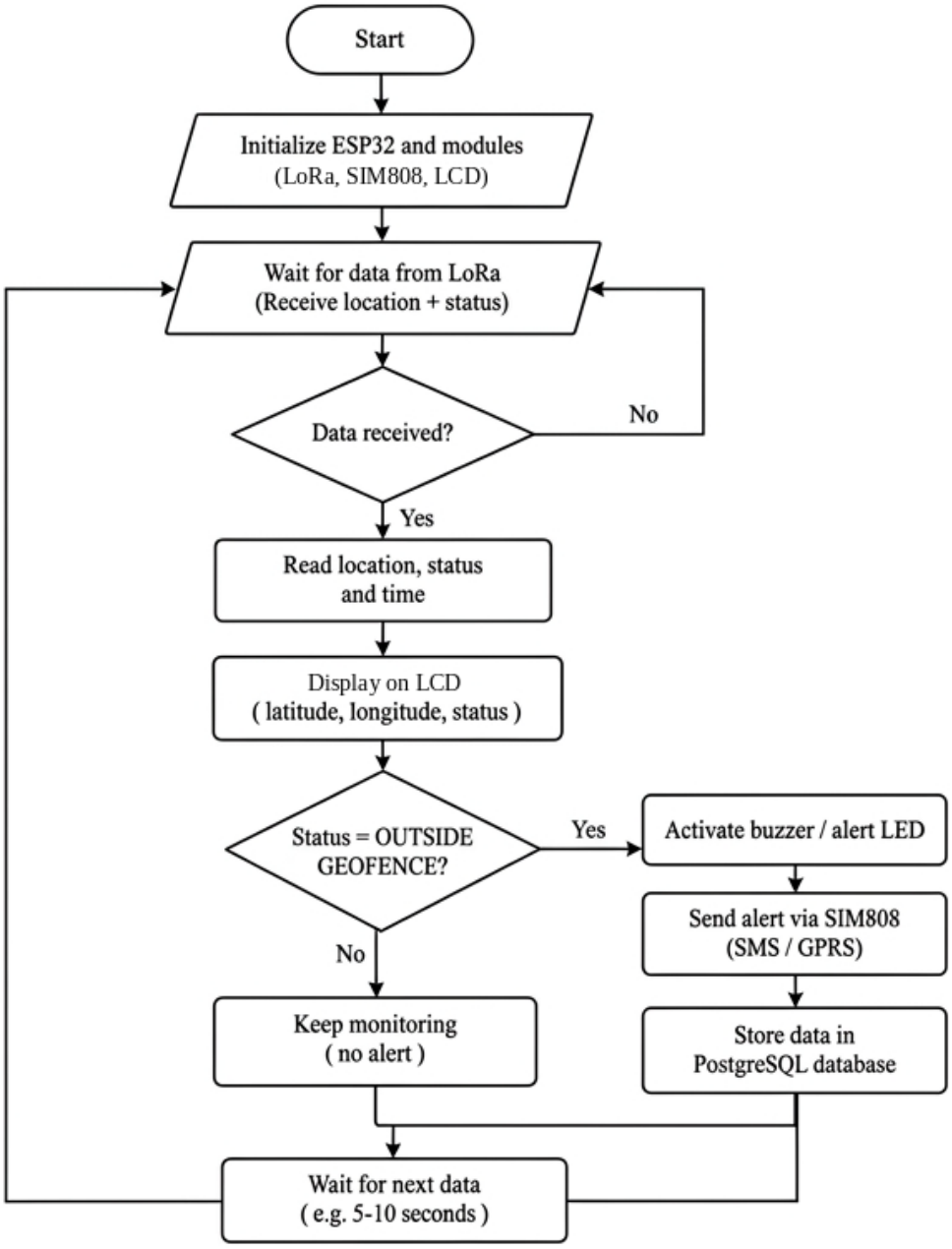


Figure 3.10: Flowchart of Monitoring Station

The Monitoring Station initializes the ESP32, LoRa receiver, SIM808 GSM module, and LCD display. It continuously waits for LoRa packets containing the cattle's location and status. When data is received, it displays the location, status, and timestamp on the LCD. If the cattle is outside the 50-meter geofence, a buzzer/LED is activated, an SMS/GPRS alert is sent, and the incident is stored in the PostgreSQL database. If the cattle remains inside the geofence, the system continues normal monitoring and repeats the process every 5–10 seconds.

CHAPTER 4: WORK PROGRESS

The work progress of the GPS-Based Geo-Fencing and Tracking System for Livestock Management has been carried out in several stages. The major hardware components have been identified and the basic system architecture has been developed. The ESP32, NEO-6M GPS, LoRa modules, SIM808 GSM module, LCD display, and alert components have been selected for the proposed system.

The GPS module has been tested with the ESP32 to obtain latitude and longitude coordinates. The basic geo-fencing concept has also been implemented, where the cattle's position is compared with a predefined 50-meter radius to determine whether it is inside or outside the safe zone.

Initial testing of the LoRa communication system has also been carried out to establish communication between the transmitter and receiver. The monitoring station is being designed to receive the cattle's location and status and display the information through the LCD.

Currently, the LoRa communication, GSM alert system, and PostgreSQL database integration are being developed and tested. Further work will focus on integrating all the modules, improving communication reliability, testing the complete system under real conditions, and conducting field testing with cattle.

4.1 Circuit Testing

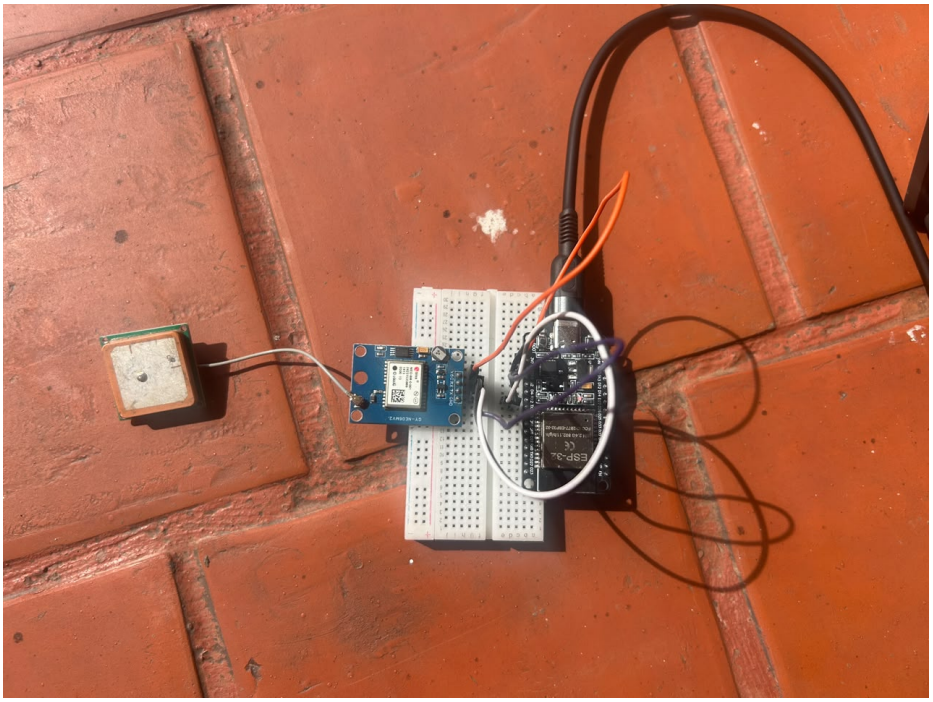


Figure 4.1: Circuit testing for GPS-Module

For testing the GPS module, the NEO-6M GPS module is connected to the ESP32 DevKit V1 using four connections. The VCC pin of the GPS module is connected to the 5V/VIN pin of the ESP32 for power supply, while the GND pin is connected to the GND pin of the ESP32 to provide a common ground. The TX pin of the NEO-6M is connected to GPIO 16 (RX2) of the ESP32, and the RX pin is connected to GPIO 17 (TX2) for serial communication. The ESP32 uses its UART2 interface at a baud rate of 9600 bps to receive GPS data. During testing, the GPS module is placed in an open outdoor area with a clear view of the sky to obtain signals from GPS satellites. The received latitude, longitude, altitude, and number of satellites are displayed on the Arduino Serial Monitor.

Table 4.1: NEO-6M GPS and ESP32 Pin Configuration

NEO-6M GPS Pin	**ESP32 Pin**	**Function**
VCC	3.3V/VIN	Power Supply
GND	GND	Ground
TX	GPIO 16 (RX2)	GPS Data Transmission
RX	GPIO 17 (TX2)	GPS Data Reception

4.2 GEO-Fence location allocation

The geo-fence location was determined using the NEO-6M GPS module connected to the ESP32 DevKit V1. The GPS module was placed at the desired livestock monitoring location in an open area, and the latitude and longitude obtained from the Serial Monitor were recorded as the center coordinates of the geo-fence. A circular boundary of 50 meters radius was then defined around this center point in the ESP32 program. The GPS module was moved to different locations to test the boundary. If the calculated distance from the center was within 50 meters, the location was considered inside the geo-fence; otherwise, it was considered outside.

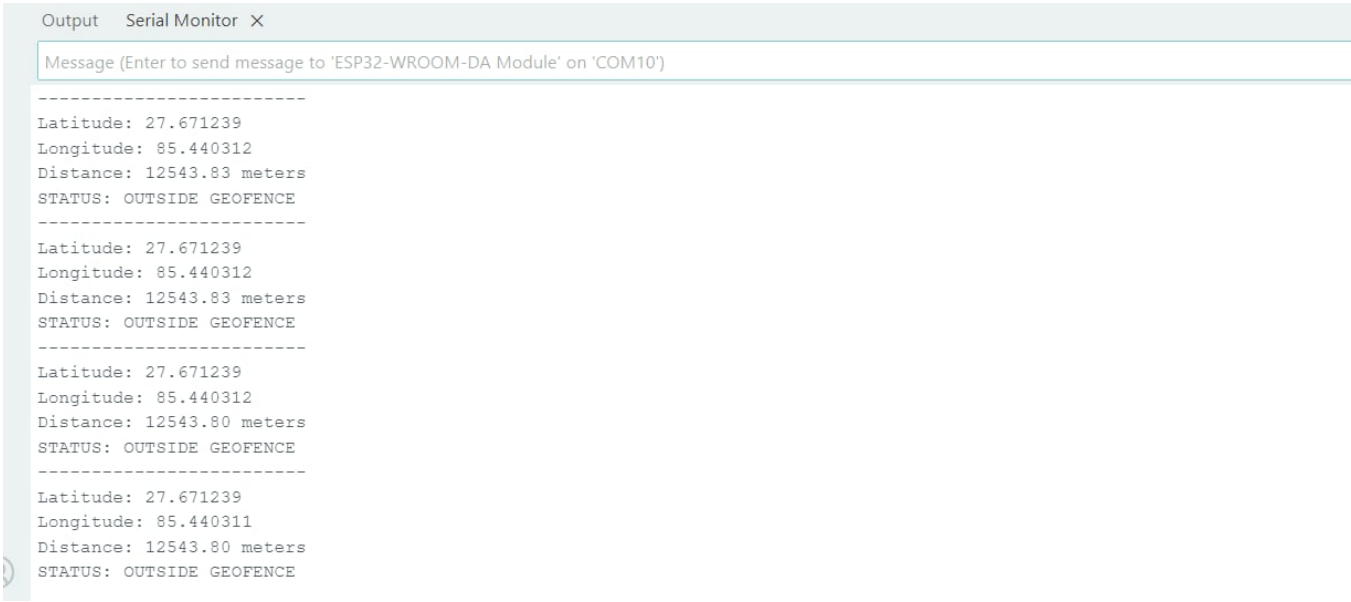


Figure 4.2: Output for the fencing allocation

4.3 Current Status

Table 4.2: Current Work Progress of the System

Module	**Status**
ESP32	Completed
GPS	Tested
Geo-fencing	Implemented
LoRa	Testing
LCD	Testing
GSM/SIM808	Testing
PostgreSQL	Developing
Complete Integration	In Progress

CHAPTER 5: FUTURE WORKS

1. LoRa Communication: The LoRa communication system will be integrated and tested for long-range data transmission. It will be used to send livestock location and geo-fence status to the monitoring station.

2. GSM Communication: The GSM module will be integrated to provide location and geo-fence alerts through SMS. This will provide an alternative communication method in areas without LoRa connectivity.

3. Database Development: A database will be developed to store livestock information, GPS coordinates, and movement history. Geo-fence violation records will also be stored for future monitoring.

4. Monitoring Dashboard: A web-based dashboard will be developed to display the livestock's current location and geo-fence status. It will help users monitor livestock remotely.

5. Field Testing: The complete system will be tested with livestock in real outdoor conditions. The testing will evaluate GPS accuracy, communication range, and overall system reliability.

CHAPTER 6: EXPECTED RESULTS

6.1 Successful System Initialization

The system is expected to initialize all hardware components — the ESP32 microcontroller, GPS module, and LoRa/GSM communication module — without errors upon power-up. All modules should establish proper communication within a few seconds of startup, confirming that the system is ready for real-time monitoring operations.

6.2 Accurate GPS Coordinate Acquisition

The GPS module is expected to acquire satellite signals and provide accurate latitude and longitude coordinates of the tracked object or individual. The location data should be updated at regular intervals, typically every few seconds, ensuring continuous and reliable positional awareness throughout the monitoring period.

6.3 Reliable Geo-Fence Detection

The system is expected to correctly and consistently determine whether the tracked entity is inside or outside the defined geo-fence boundary. The comparison algorithm running on the ESP32 should produce accurate results with minimal false positives or false negatives, ensuring that boundary breaches are detected precisely and promptly.

6.4 Real-Time Dashboard Updates

When the tracked entity remains within the geo-fence, the system is expected to continuously update the monitoring dashboard with the current location in real time. The dashboard should reflect position changes smoothly, allowing authorized users or administrators to visually track the subject's movement within the safe zone at any given moment.

6.5 Immediate Alert Generation on Boundary Breach

Upon detection of a geo-fence violation, the system is expected to instantly activate the onboard buzzer and LED indicator as a local alert. This response should occur with minimal delay, ideally within one monitoring cycle, to ensure that the breach is flagged as soon as it is detected without any significant lag in the physical alerting mechanism.

6.6 Successful Remote Notification via LoRa/GSM

Following a boundary breach, the system is expected to successfully transmit an alert message to the designated recipient through either the LoRa or GSM communication module. The notification should include relevant information such as the time of breach and the last known coordinates, enabling remote users to respond quickly regardless of their physical location.

6.7 Consistent Data Logging in Cloud/Database

All location data and event records — including both normal updates and breach events — are expected to be stored accurately and consistently in the cloud database. The stored data should be accessible for review, allowing administrators to analyze movement history, identify patterns, and retrieve records of any past geo-fence violations for reporting or security purposes.

6.8 Continuous and Uninterrupted Monitoring Loop

The overall system is expected to operate in a stable and continuous loop without crashing, freezing, or losing connectivity during extended operation. Each cycle of reading GPS coordinates, performing geo-fence comparison, triggering appropriate responses, and storing data should execute reliably, demonstrating the system's robustness and suitability for long-term real-world deployment.

REFERENCES

- Kaplan, E. D., & Hegarty, C. J. (2006). *Understanding GPS: Principles and applications* (2nd ed.). Artech House.
- Li, Y., Zhuang, Y., Hu, X., Gao, Z., Hu, J., Chen, L., & Thompson, J. (2020). *Location-enabled IoT: A survey of positioning techniques, error sources, and mitigation*. IEEE Internet of Things Journal.
- Khan, M. A., Saboor, A., Kim, H. C., & Park, H. (2021). *A systematic review of location aware schemes in the Internet of Things*. Sensors, 21(9).
- Bonilla, V., Campoverde, B., & Yoo, S. G. (2023). *A systematic literature review of LoRaWAN: Sensors and applications*. Sensors, 23(20).
- Shahbazian, R., Macrina, G., Scalzo, E., & Guerriero, F. (2023). *Machine learning assists IoT localization: A review of current challenges and future trends*. Sensors, 23(7), 3551.
- Tobin, K., Heidari, O., Volpi, C., Sodder, S., & Duncan, D. (2023). *Use of geofencing interventions in population health research: A scoping review*. BMJ Open, 13(8), e069374.
- Reddy, K. D. P., Vamshi, K., Chowdary, K. M. H., Prakash, M., Rehman, S. A., & Sumaiya, M. N. (2024). *IoT-based campus monitoring using LoRa technology: A review*. Journal of IoT-based Distributed Sensor Networks.

COST ESTIMATION

Table 6.1: Estimated Cost of the Proposed System

S.N.	Component	Quantity	Unit Cost (NPR)	Total Cost (NPR)
1	ESP32	1	1,200	1,200
2	NEO-6M GPS Module	1	1,500	1,500
3	LoRa SX1278 Module	2	1,500	3,000
4	LCD Display	1	500	500
5	Buzzer	1	300	300
6	Li-ion Battery	1	800	800
7	Battery Charging Module	1	500	500
8	Connecting Wires	1 Set	200	200
9	Breadboard / PCB	1	300	300
10	Sensors	1	1000	1000
11	Enclosure / Casing	1	500	500
12	Miscellaneous	-	2000	2000
			Total Estimated Cost	**11,800**

GANTT CHART

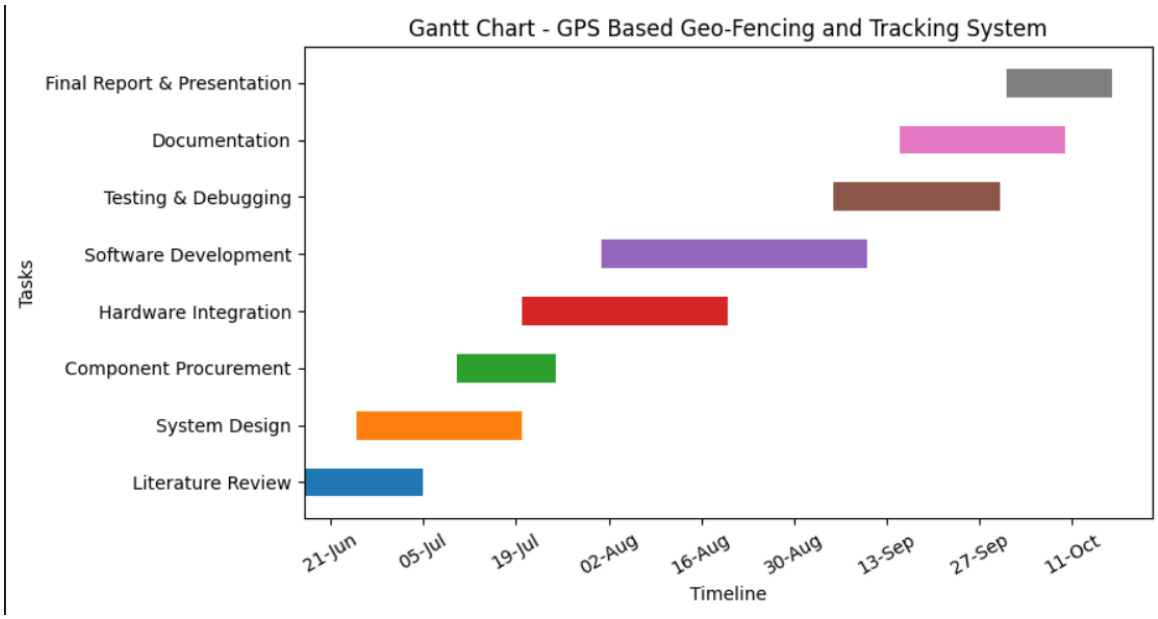


Figure 6.1: Gantt chart